

INSTRUCTION MANUAL

ROTARY VACUUM PUMPS 'ED' SERIES



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INTRODUCTION

a. General Description

'HINDHIVAC' Direct Driven, Vane type vacuum pumps are double stage, oil sealed type and designed for a variety of basic vacuum pumping applications.

The pumps are similar in construction but of varying pumping speeds (nominal free air displacement) like $3(\text{M}^{-3})/\text{hr}$, $6(\text{M}^{-3})/\text{hr}$, $15(\text{M}^{-3})/\text{hr}$, $21(\text{M}^{-3})/\text{hr}$ and $30(\text{M}^{-3})/\text{hr}$. The direct drive to the pump is provided through a flexible coupling from a flange mounted motor which also drives the cooling fan. The pumps are provided with either single phase or three phase motors.

b. Construction

'HINDHIVAC' High Vacuum Pumps, ED-Series, are spring loaded Sliding Vane type, with vanes placed in the slots of the rotor. They are mounted eccentrically both in the first and second stage with inter connecting ports.

Both, the first stage which creates primary vacuum and second stage which creates the low pressure, are isolated with the introduction of an isolator in between and the two rotors are mounted eccentrically within respective stators.

The first stage end plate has a bearing and an oil seal for isolation from atmosphere. The second stage end cover also has a bearing for locating the shaft, which will give the closest possible tolerance and free movement within the stator and rotor for efficient performance of the pump.

This block is fixed to an aluminum assembly, called the Mounting Block. On one side of this is fixed the vacuum pump and the oil cover and on the other side the driving motor. A coupling joins this assembly to the motor for a smooth drive.

The oil pump housing is mounted on the rear end plate of the stator. This oil pump provides the lubrication to the pump and the oil flow to the stator. The pump functions in a similar manner as the vacuum pump. It has a vane, mounted in an eccentric position in the bore and drives the oil under pressure. In the event of the pump stopping a spring loaded flap valve is activated to seal the oil port of the oil pump, thereby preventing any back-streaming of oil into the chamber because of difference in pressure.

A filter is provided on the oil pump housing through which oil is sucked in. This prevents any dirt or fibrous tissues from entering the pump and causing reduction in flow of oil and thereby seizure of moving parts.

Precautions should also be taken when these pumps are used in places where high water vapour tolerance is required. Chilled condensers are recommended for condensing water vapours. While handling water vapour, please run the pump with its gas ballast open.

Special care is to be taken when pumps are used where acidic vapours are present. Traps to neutralize these dangerous vapours must be used. Otherwise, they will damage the inner parts of the pumps.

For smooth functioning and noiseless operation, epoxy blades are used within the rotors. Chemicals prone to attack such resins should be specially taken care of and should not be allowed inside the pump. Otherwise, decomposition of the vane will result in unsatisfactory functioning of the pump.

Occasional opening of oil tank and cleaning the filters is recommended to give trouble free service and assured low pressures.

c. Working Principle

During operation, the rotor vanes sweep the volume of the gas or air trapped in the crescent shaped gap formed by the rotor which is mounted eccentrically in the stator. As each van passes the inlet port opening a known quantity of gas is introduced and

subsequently trapped and compressed by the next vane following it and ejected via and exhaust flap valve mostly and via the interconnecting port to the II stage partially, when the inlet pressure is near atmospheric pressure. As the inlet pressure drops the I stage exhaust flap valve closes and all the air or gases pass to the II stage, where it is further compressed and discharged to atmosphere.

d. Gas Ballast

The Direct drive models are incorporated with a gas ballast facility to enable them to pump the condensable vapours without contaminating the pump oil. This is done by introduction of gas or air at atmospheric pressure through a manually operated valve into the volume between the second stage rotor vane and discharge valve, when the mixture of air and vapour in this volume is at low pressure. When the volume of gas (air)/vapour mixture is compressed prior to ejection, the discharge valve opens before the partial pressure of the vapour component is high enough to cause it to condense.

TECHNICAL SPECIFICATIONS

Sl. No.	Pump Model	ED-6	ED-15	ED-21	ED-30
1	Nominal Pumping Speed: M ³ /Hr Lit/Min	6 100	15 250	21 350	30 500
2	Ultimate Partial Pressure: (on Mcleod Gauge) Gas Ballast Closed: (m.bar) Gas Ballast Opened: (m.bar)	5×10^{-4} 6×10^{-2}	5×10^{-4} 6×10^{-2}	5×10^{-4} 6×10^{-2}	5×10^{-4} 6×10^{-2}
3	Weight (kgs)	32	42	45	52
4	Maximum nominal power rating (KW)	0.25	0.37	0.55	1.1
5	Pump Rotational Speed (at no load r.p.m)	1340-1440	1340-1440	1340-1440	1340-1440
6	Oil Capacity (Lits)	2.0	2.75	2.75	3.0
7	Inlet Flange (ISO-KF)	KF-25	KF-25	KF-25	KF-25
8	Recommended Oil	HINDHIVAC Molecular Distilled Oil: Grade MD-504			

INSTALLATION

a. Unpacking

Unpack the pump and remove all protective covers and the protective cover on the inlet pipe. Check for any possible damage in transit. If found alright, proceed as given below. The handle of the pump which is packed separately is to be fixed with the two screws provided at the top of the pump.

b. Pump Inlet Connection

The vacuum inlet connection is a KF-25 flange with an 'O' ring holder and 'O' ring. The free flange can be welded to a pipe or inserted into a rubber tubing.

c. Pump Outlet Connection

A plain nozzle of suitable size provided for the pump discharge connection.

The pump exhaust nozzle can be connected through a tubing to the outside of the building with a sump in such a way that condensed fluid from pipe should not flow back to the pump.

d. Oil Charge

During installation, initially pour 100ml of oil to the inlet of the pump to facilitate oil pumping. Remove the oil filling plug and fill oil to a level which is visible through the sight glass using the recommended grade of oil. The level of the oil is to be half the level of the oil sight glass. Replace the oil filling plug with the sealing gasket in position.

NOTE: During normal running of the pump, the oil level will be visible in the sight glass.

e. Electrical Connections

The pumps are fixed with single phase 230V AC 50Hz motors of suitable rating. Three phase motors are provided on request at extra cost.

To connect the electrical supply leads, remove the terminal box cover and take the lead wires through and connect to the appropriate terminals.

The direction of rotation is clockwise when viewed from drive end. The motor is factory wired to run in the correct direction and requires no change while connecting power in case of single-phase motors.

NOTE: Never obstruct the pump outlet, this may cause dangerous pressure.

- a. Before starting the pump check that the oil level, which is visible in the oil sight glass, is at the indicated level. The pump will operate satisfactorily at this level for normal use.
- b. When pumping condensable vapours present in the system, the gas ballast facility must be utilized by introducing air into the pump by rotating the gas ballast valve in anti-clockwise direction. Before pumping vapours, the pump should be isolated from the system and allowed to run for approximately 20-30 minutes to warm the oil and assist in preventing vapour condensation in the cold pump. The gas ballast valve is to be closed when better ultimate vacuum (lower pressure) is required in the system after the vapours are removed.
- c. When the pump is used for pumping large quantities of vapour, it is recommended that the pump is isolated from the system and run for atleast 30 minutes with the gas ballast valve open. This particularly applies after corrosive vapours, have been pumped. If the pump is used in systems which produce heavy evacuation of vapours, particularly if vapours are of a corrosive nature, the gas ballast may not completely protect the pump. A suitable trap is recommended to reduce the harmful vapours carried over to the pump.
- d. If the system connected to the pump has likelihood of solid particles present entering the pump, a dust filter should be incorporated in the line. (for details refer to accessories).
- e. When measuring ultimate pressure (ultimate vacuum) it is to be taken note of, that the pump oil may have a vapour pressure which is higher than that of the permanent

gases in the system being evacuated. Hence, while measuring the ultimate vacuum with gauges of Pirani or Thermocouple type, which measure the total pressure of the permanent gases and vapours, they will indicate differently from gauges of McLeod or Manometer type which will only indicate the pressure of the permanent gases.

To obtain the highest possible vacuum when the pump is filled with a fresh charge of oil, it is recommended to run the pump with full gas ballast open for 10 to 15 minutes before measuring the ultimate vacuum or connecting it to the system. This has the effect of degassing the oil and improving the ultimate vacuum.

- f. The retention of vacuum in the system when the pump is shut off for any reason is possible only if the gas ballast valve is closed. It is recommended to use a Solenoid Operated Isolation Cum-Air Admittance Valve at the inlet of the pump where greater system security is required, on large volumes which are evacuated where they create a negative drive on the pump by force of a large volume.
- g. The pump is designed for long and trouble free operation provided the recommended operating and servicing procedures are adhered to. Before dispatching the pump, each and every pump is inspected and tested for performance. Many cases of suspected failure and of poor pump performance are in fact due to leakage in vacuum system, faulty due to wrong selection or not properly providing suitable accessories like traps, etc. Any such possibilities are to be checked with proper guidance by our design department while choosing pumps for different applications.

MAINTENANCE

The following routine servicing will be required:

- a. **Oil Level**

Oil level to be checked daily.

- b. **Oil Pouring**

To top the oil, remove the oil filter plug located on the top of the oil tank and fill with recommended grade of oil until the oil level is visible in the sight glass above the

minimum required level. Replace the filler plug and ensure the gasket seal is in position.

c. Oil Filing

To change the oil proceeds as follows:

d. Gravity Drain

Switch off the pump, place a clean suitable tray or container beneath the drain plug and remove the plug. Allow oil to drain completely. Switch on the pump for few seconds and throttle the exhaust nozzle with finger to ensure complete flushing of the oil. Switch off the pump immediately. Pour a small quantity of oil down the vacuum inlet and switch on the pump for rinsing the pump and flushing purpose. Switch off the pump. Drain the oil and replace the drain plug. Remove the oil filling plug and refill the pump to the correct level with the recommended grade of oil. Replace the filler plug and ensure the gasket seal is in position.

	W A R N I N G	DO NOT completely restrict the pump outlet while flushing the oil under pressure as this may cause excessive high internal Pressure build up resulting in pump oil tank rupture. If long pipes are connected to the pump exhausting out of air conditioned environments, please use sufficiently large diameter pipes to ensure no back pressure is built up on exhaust.
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SPARES & ACCESSORIES

A set of spares comprising of these items are recommended for trouble free maintenance.

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|----|--|---|-------|
| a. | Springs & Pins for Vanes (1 st & 2 nd Stage) | : | 1 set |
| b. | Oil Seals | : | 1 set |
| c. | Gaskets and 'O' Rings | : | 1 set |
| d. | Filters | : | 1 set |
| e. | 'O' ring holder with 'O' Ring | : | 1 set |
| f. | K.F. Clamp | : | 1 set |

A full range of accessories are available for direct drive pumps as listed below.

a. Inlet Filters (Dust Filters)

Inlet dust filters are recommended where fine abrasive dust particles are to be prevented from entering the vacuum pump. This filter incorporates a fine mesh filter which is supported by a frame. The inlet and outlet are of KF couplings.

b. Moisture Trap

The moisture trap is designed for use in systems where the requirement is for removal of limited quantities of moisture at low pressures. The phosphorous pentoxide (P₂O₅) desiccant is contained within the trap.

c. Foreline Trap

For use in clean pumping systems, it utilizes the trapping properties of activated alumina to trap oil vapour and prevent back migration of pump oil vapour into the vacuum system.

d. Mist Filter

The Exhaust Mist Filter is recommended on the outlet (exhaust) to capture oil mist which would be otherwise admitted into the environment while running on gas ballast or inlet open for long periods of time.

e. Anti Vibration Mounting

'U' shaped rubber cushion mounting for pumps installed on framed structures to avoid vibrations being transmitted to other parts. Flexible hoses or bellows between the pump and the system are recommended.

f. Isolation Valve

The Isolation valve is recommended for enhanced protection against back streaming of vacuum pump oil into the system. This is incorporated directly on the inlet connection of the pump.

g. Water Cooled Condenser

The water cooled condenser is recommended where condensable vapours or gases are pumped by the pumping which may otherwise contaminate the pump oil. These are supplied with KF couplings to match the pump inlet.

WARRANTY CONDITIONS

Warranty of one year for any manufacturing or material defects, faulty workmanship from the date despatch of the pump.

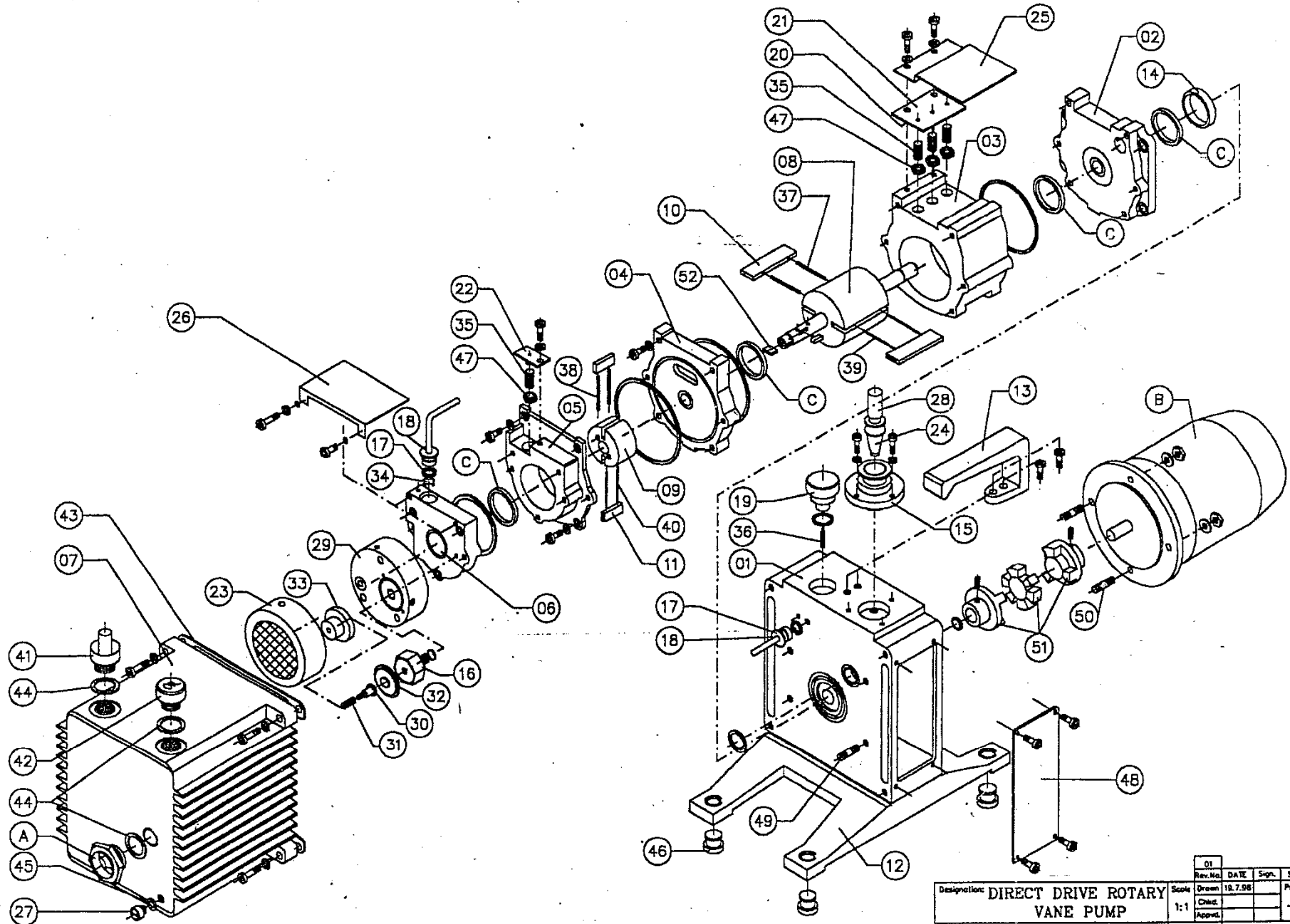
If any fault occurs or parts are damaged contact the service department of

HINDHIVAC PRIVATE LTD.
BANGALORE
Or
Regional service centers
At
MUMBAI, DELHI, CHENNAI, HYDERABAD, PUNE,
AHMEDABAD AND CALCUTTA

Specifying:

1. Model No. of the Pump.
2. Serial No. of the Pump.
3. Date of Purchase.
4. Nature of fault.

The service department will normally arrange for service personnel to call or for the faulty pump to be returned. No pump should be returned without prior agreement. The above procedure also applies to pumps damaged during transit, but both the carrier and Insurance Company on which the goods are insured to be notified within three days of receipt of goods.



		01			
		Rev. No.	DATE	Sign.	SHEET 1/2
Designation:	DIRECT DRIVE ROTARY VANE PUMP	Scale	Drawn	19.7.98	Projection 
		1:1	Checked		
		Approved			
		HINDHVAC PRIVATE LTD BANGALORE			
				ED-SERIES-EX	

ITEM NO.	DESCRIPTION	QTY
01	MOUNTING BLOCK	1
02	END COVER MOUNTING BLOCK SIDE	1
03	I STAGE STATOR	1
04	ISOLATOR	1
05	II STAGE STATOR	1
06	END COVER	1
07	OIL TANK	1
08	I STAGE ROTOR	1
09	II STAGE ROTOR	1
10	I STAGE VANE	2
11	II STAGE VANE	2
12	BASE PLATE	1
13	HANDLE	1
14	LOCATING BUSH	1
15	INLET CONNECTION	1
16	OIL PUMP HOUSING BUSH	1
17	G.B.CONNECTING PLUG	2
18	G.B.CONNECTING PLUG HOLDER	2
19	GAS BALLAST KNOB	1
20	GUIDE PIN	4
21	I STAGE EXHAUST PLATE	1
22	II STAGE EXHAUST PLATE	1
23	OIL FILTER	1
24	AIR INLET FILTER	1
25	I STAGE BAFFLE COVER	1
26	II SGATE BAFFLE COVER	1
27	OIL TANK DRAIN PLUG	1
28	INLET PIPE	1
29	OIL PUMP HOUSING	1
30	OIL PUMP HOUSING SPACER	1

ITEM NO.	DESCRIPTION	QTY
31	OIL PUMP HOUSING SPRING	1
32	OIL PUMP HOUSING FLAP	1
33	OIL PUMP HOUSING TOP COVER	1
34	FLOAT VALVE	1
35	I & II STAGE EXHAUST POPPET SPRING	4
36	GAS BALLAST SPRING	1
37	I STAGE VANE SPRING	2
38	II STAGE VANE SPRING	2
39	I STAGE VANE SPRING PIN	2
40	II STAGE VANE SPRING PIN	2
41	EXHAUST NOZZLE	1
42	OIL POURING PLUG	1
43	OIL TANK GASKET	1
44	WASHER (FOR OIL POURING PLUG & EXHAUST NOZZLE)	2
45	DRAIN PLUG WASHER	1
46	RUBBER GROMMET	4
47	EXHAUST POPPET	4
48	NAME PLATE	2
49	STUD (FOR END COVER MOUNTING BLOCK SIDE)	4
50	STUD (FOR MOTOR)	4
51	FLEXIBLE COUPLING SET	1
52	OIL PUMP VANE	1
A	OIL SIGHT GLASS	1
B	MOTOR	1
C	OIL SEAL	4

Designation: DIRECT DRIVE ROTARY VANE PUMP	Scale	01	Rev.No.	DATE	Sign.	SHEET	2/2
	1:1	Drawn	19.7.98			Projection 	
		Chkd.					
		Appvd.					
HINDHIVAC PRIVATE LTD BANGALORE			ED-SERIES-EX				